P1 - Energy Booklet (HT)

2. What are the 8 energy stores?

	Energy Store	Example
	Kinetic energy	Moving car
	Gravitational potential energy	An object at height (book on a shelf)
	Chemical energy	Muscles, food, battery, fuel
	Thermal energy	Anything hot (bonfire)
	Elastic potential energy	Stretched or compressed object
	Nuclear energy	The Sun, a nuclear plant
	Magnetic energy	Magnets near each other
	Electrostatic energy	Charges interacting with each other

3. What is the principle of conservation of energy?

Energy cannot be created or destroyed, only stored or transferred.

4. What are the 4 energy transfer pathways?

Mechanical (work)
Electrical
Waves - sound and light
Heating

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5. Define “work done”.

Energy transferred when a force is used to move an object over a distance.

6. What formula relates work done, force and distance?

Work done = Force × Distance

7. Define “power”.

The rate (per second) of work done: how much energy is transferred each second.

8. What are the units of power?

Watts (joules per second).

9. What formula relates work done, power and time?

Work done = Power × time
or **P = E / t**

10. What is meant by “1 kilowatt hour”?

The energy used by something with a power of 1 kW (1,000 W) in a time of 1 hour.

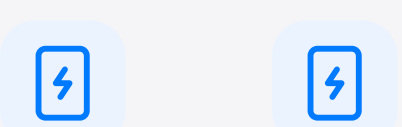
Q5–6 · Work done



W = F × d

Q7–9 · Power

Same energy: 600 J



30 W — 20 s

60 W — 10 s

P = E / t

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11. The kinetic energy stored by an object depends on what 2 things?

The mass of the object (m).
The speed the object is moving at (v).

12. What formula relates kinetic energy, mass and speed?

Kinetic energy = ½ × m × v²

13. The gravitational potential energy stored by an object depends on what 3 things?

The mass of an object (m).
The height the object is at (h).
The gravitational field strength around the object (g).

14. What formula relates gravitational potential energy, mass, height and gravitational field strength?

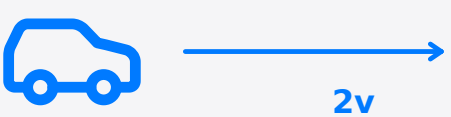
E_p = mass × gravitational field strength × height
E_p = m × g × h

Q11–12 · Kinetic energy

Same mass (m)



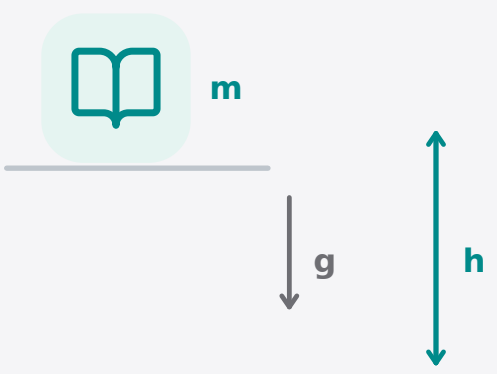
Kinetic energy E_k



4 × kinetic energy

v^2 = speed × speed

Q13–14 · Gravitational potential energy



E_p = m × g × h

15. What is meant by “dissipation”?

Energy wasted during an energy transfer (almost always transferred to the thermal energy store of the surroundings).

16. Define “efficiency”.

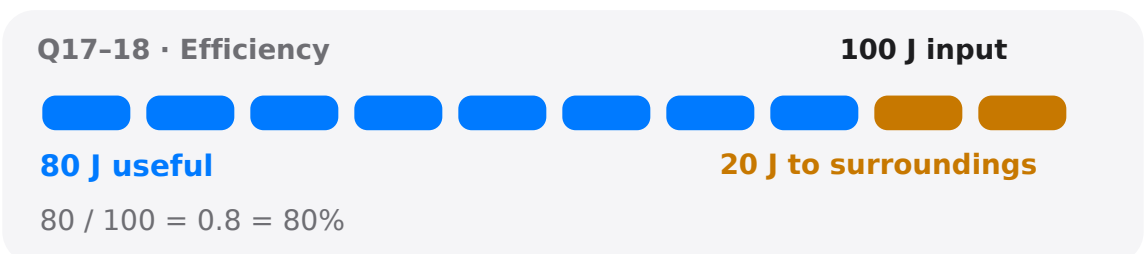
The proportion/percentage of energy that is usefully transferred in a process.

17. What is the formula for efficiency?

Efficiency = useful energy output / total energy input

18. What is the maximum possible efficiency? (As a percentage and as a number.)

100% or 1.



19. Should insulation have a high or low thermal conductivity?

Low.
The lower it is, the less heat it will conduct.

20. Name three ways of reducing unwanted energy transfers (wasted energy).

Insulation
Lubrication
Streamlining / aerodynamics

21. What are the fossil fuels?

Coal, (crude) oil and (natural) gas.

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22. What non-renewable energy source is not a fossil fuel?

Nuclear.

23. What greenhouse gases are released when fossil fuels are burned?

Carbon dioxide
Sulphur dioxide
Nitrogen oxide
(Methane is released during coal mining.)
You need to know more than just CO_2.

24. What does “non-renewable” mean?

It cannot be replenished / it will run out.

25. What does “renewable” mean?

It can be replenished / it will not run out.

26. Name 3 renewable energy sources.

Solar, wind, geothermal, hydroelectric, tidal or biomass.

CO_2 and methane are greenhouse gases; sulphur dioxide and nitrogen oxides are pollutants associated with acid rain.



Solar



Wind



Geothermal



Hydroelectric



Tidal



Biomass

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Name	Symbol	Factor	To convert to standard units:
Giga	G	10^{9}	× 1,000,000,000
Mega	M	10^{6}	× 1,000,000
kilo	k	10^{3}	× 1,000
Centi	c	10^{-2}	÷ 100
milli	m	10^{-3}	÷ 1,000
micro	μ	10^{-6}	÷ 1,000,000
nano	n	10^{-9}	÷ 1,000,000,000

Symbol	Quantity	Definition	Unit
E	Energy	Can be stored and transferred between different stores.	Joules (J)

× 1,000

1 kW → **1,000 W**

÷ 100

100 cm → **1 m**

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Symbol	Quantity	Definition	Unit
F	Force	A resultant force changes the speed, direction or shape of an object.	Newtons (N)
d	Distance	How far something moves.	Metres (m)
W	Work Done	The energy required for a force to move an object over a distance.	Joules (J)
P	Power	Rate of energy transfer (energy transferred per second).	Watts (W)
t	Time	A measure of how long something takes.	Seconds (s)
m	Mass	The amount of matter in an object.	Kilograms (kg)
g	Gravitational Field Strength	Gravitational force per kilogram of mass (10 N/kg on Earth).	N/kg
h	Height	Vertical height of an object.	Metres (m)
v	Velocity	How fast something is moving in a given direction.	m/s

Formulae

Work Done	$W = F \times d$
Power	$E = P \times t$ or $P = E / t$
Gravitational Potential Energy	$E_p = m \times g \times h$
Kinetic Energy	$E_k = ½ \times m \times v^2$
Efficiency	e = useful energy output / total energy input

$$E_k = ½ \times m \times v^2$$

m: mass in kg

v^2: speed × speed

$$E_p = m \times g \times h$$

h: vertical height in metres